

Field:

A commutative ring $(F, +, \cdot)$ with unity ($1 \neq 0$) is called a **field** if every non-zero element of F has a multiplicative inverse in F .

i.e., $\forall a \in F \setminus \{0\}, \exists a^{-1} \in F$ such that:

$$a \cdot a^{-1} = a^{-1} \cdot a = 1$$

In other words, $(F, +, \cdot)$ is a field if:

1. $(F, +)$ is an abelian group
2. $(F \setminus \{0\}, \cdot)$ is an abelian group
3. Distributive laws hold

Examples of Fields:

1. $(\mathbb{Q}, +, \cdot), (\mathbb{R}, +, \cdot), (\mathbb{C}, +, \cdot)$ are fields.
2. $(\mathbb{Z}_p, +_p, \times_p)$ is a field if and only if p is prime.

Non-examples:

1. $(\mathbb{Z}, +, \cdot)$ is not a field (only ± 1 have multiplicative inverses).
2. $(\mathbb{Z}_n, +_n, \times_n)$ where n is composite is not a field.

Skew-Field / Division Ring:

A ring $(R, +, \cdot)$ with unity ($1 \neq 0$) is called a **division ring** (or **skew-field**) if every non-zero element has a multiplicative inverse in R (multiplication need not be commutative).

Theorem:

Every field is an integral domain.

Proof:

Let F be a field. Then F is a commutative ring with unity.

To show F is an integral domain, we must show it has no zero divisors.

Let $a, b \in F$ such that:

$$ab = 0$$

If $a = 0$, then there is nothing to prove.

If $a \neq 0$, then since F is a field, $a^{-1} \in F$ exists.

Multiplying both sides by a^{-1} :

$$a^{-1}(ab) = a^{-1} \cdot 0$$

$$(a^{-1}a)b = 0$$

$$1 \cdot b = 0$$

$$b = 0$$

Thus, $ab = 0 \implies a = 0$ or $b = 0$.

Hence, F has no zero divisors, so F is an integral domain.

Remark:

The converse of the above theorem is not true in general.
 For example, $(\mathbb{Z}, +, \cdot)$ is an integral domain but not a field.

Theorem:

Every finite integral domain is a field.

Proof:

Let D be a finite integral domain.

Let $D = \{x_1, x_2, \dots, x_n\}$ where all elements are distinct.

Let $a \in D$ such that $a \neq 0$.

Consider the set:

$$aD = \{ax_1, ax_2, \dots, ax_n\}$$

Since D is closed under multiplication, $aD \subseteq D$.

Now we show all elements of aD are distinct:

Let $ax_i = ax_j$ for $i \neq j$.

Since D is an integral domain and $a \neq 0$, by cancellation law:

$$x_i = x_j$$

which is a contradiction since elements of D are distinct.

Thus, aD contains n distinct elements of D .

Since D has exactly n elements, we have:

$$aD = D$$

Since $1 \in D$ (as D has unity), $\exists x_k \in D$ such that:

$$ax_k = 1$$

Since D is commutative:

$$ax_k = x_k a = 1$$

$$\Rightarrow x_k = a^{-1}.$$

Thus, every non-zero element of D has a multiplicative inverse in D .

Hence, D is a field.

Corollary:

$(\mathbb{Z}_p, +_p, \times_p)$ is a field if and only if p is a prime.

Characteristic of a Ring:

Let $(R, +, \cdot)$ be a ring. If there exists a least positive integer n such that:

$$n \cdot a = \underbrace{a + a + \cdots + a}_{n \text{ times}} = 0 \quad \forall a \in R$$

then n is called the **characteristic** of the ring R , denoted as $\text{char}(R) = n$.

If no such positive integer exists, then R is said to be of **characteristic zero** (or infinite).

Examples:

1. $\text{char}(\mathbb{Z}) = 0$, $\text{char}(\mathbb{Q}) = 0$, $\text{char}(\mathbb{R}) = 0$, $\text{char}(\mathbb{C}) = 0$.
2. $\text{char}(\mathbb{Z}_n) = n$.
3. $\text{char}(\mathbb{Z}_p) = p$.

Theorem:

Let R be a ring with unity 1.

1. If the order of 1 under addition is finite, say n , then $\text{char}(R) = n$.
2. If the order of 1 under addition is infinite, then $\text{char}(R) = 0$.

Proof:

1) Let $O(1) = n$ in $(R, +)$.

Then $n \cdot 1 = 0$, and n is the least positive integer with this property.

For any $a \in R$:

$$n \cdot a = a + a + \cdots + a = (1 + 1 + \cdots + 1)a = (n \cdot 1)a = 0 \cdot a = 0$$

Thus, $n \cdot a = 0$ for all $a \in R$.

Since n is the least positive integer for which $n \cdot 1 = 0$, it is also the least positive integer for all elements.

$\therefore \text{char}(R) = n$.

2) If $O(1)$ is infinite, then there is no positive integer n such that $n \cdot 1 = 0$.

Hence, no such n exists for all elements, so $\text{char}(R) = 0$.

Theorem:

The characteristic of an integral domain (or a field) is either 0 or a prime number.

Proof:

Let D be an integral domain. If $\text{char}(D) = 0$, then we are done.

Suppose $\text{char}(D) = n > 0$. We need to show that n is a prime.

If possible, let n be composite, so $n = n_1 n_2$, where $1 < n_1, n_2 < n$.

Then:

$$\begin{aligned} n \cdot 1 = 0 &\implies (n_1 n_2) \cdot 1 = 0 \\ &\implies (n_1 \cdot 1)(n_2 \cdot 1) = 0 \end{aligned}$$

Since D is an integral domain (no zero divisors):

$$n_1 \cdot 1 = 0 \quad \text{or} \quad n_2 \cdot 1 = 0$$

This contradicts the fact that n is the least positive integer such that $n \cdot 1 = 0$ (since $n_1, n_2 < n$).

Hence, n cannot be composite.

$\therefore n$ must be a prime number.